

Review

Computational Études — A Spectral Approach

Dr. Denys Dutykh Abu Dhabi, UAE 2026

535 pp., Typst-typeset, tri-lingual code (Python / MATLAB / Julia)

May 2026

Overview

Computational Études is an ambitious, well-crafted graduate textbook on spectral methods for partial differential equations. Its 535 pages span 17 chapters plus an Afterword, covering separation of variables, polynomial interpolation, Chebyshev and Fourier differentiation matrices, boundary value problems, pseudospectral methods for nonlinear PDEs, quadrature, and time-stepping — all illustrated with three-language (Python, MATLAB, Julia) implementations and thorough literature surveys in every chapter. The scope is genuinely broad: a reader who works through the book from cover to cover will have encountered the classical analytical foundations of Chapter 2, the subtleties of node geometry and the Runge phenomenon in Chapter 4, the theoretical backbone of spectral accuracy in Chapter 6, a comprehensive survey of boundary condition strategies in Chapters 8 and 13, and a gallery of nonlinear PDE simulations — KdV solitons, Kuramoto–Sivashinsky chaos, two-dimensional Navier–Stokes vortex dynamics — in Chapters 11 and 14. The late chapters on quadrature (Chapter 15) and the periodic trapezoidal rule (Chapter 16), built around landmark papers of Trefethen and Weideman, are particularly striking: they overturn standard textbook narratives in a way that is both intellectually honest and pedagogically memorable.

The mathematical exposition is rigorous yet readable. Theorems are stated with precision, but the author never allows formalism to obscure physical intuition. Each abstract result is promptly illustrated with a concrete numerical experiment, and the five-convergence-class taxonomy that runs like a thread through Chapters 6, 15, and 16 gives the reader a lasting conceptual framework for understanding *why* spectral methods work, not merely *how* to use them. The code examples are generally correct and well-explained across all three languages. The decision to provide full, standalone implementations in Python, MATLAB, and Julia — rather than pseudocode or a single reference language — substantially increases the book’s practical value, since readers from different computational backgrounds can engage with the material without a translation step. The literature surveys, which close every chapter, are another distinctive strength: they are historically deep, thematically organised, and consistently updated to 2025–2026, so the reader finishes each chapter with both a solid foundation and a clear view of the current research frontier.

The book’s pedagogical framing as a collection of musical *études* is both apt and attractive. Like a pianist’s étude, each computational exercise is designed to isolate and develop a specific technical skill — the leapfrog time-stepper, the integrating-factor method, the Kronecker-product assembly of a 2D Laplacian — while also being musically complete in itself, producing a physically meaningful and visually engaging result. This structure makes the book unusually pleasant to read: one never has the sense of enduring preparatory material in order to reach the “real” content, because every chapter delivers something concrete and beautiful.

The issues identified in this review are, with one exception, typographical inconsistencies

and structural gaps in the table of contents: a missing accent on three instances of the word *étude*, two computational études absent from the TOC, an inconsistency in section-heading capitalisation across the second half of the book, and an unremoved editorial self-correction in one exercise. The sole mathematical error — an incorrect Bernstein ellipse parameter in §7.5.2 (value stated as 1.5, correct value is the golden ratio $(1 + \sqrt{5})/2 \approx 1.618$) — is easy to fix and does not affect the surrounding exposition once corrected.

More broadly, this book stands out for a style that is simultaneously modern and pedagogical, a combination that is rarer than it should be in the numerical analysis literature. Most introductory spectral methods texts were written in the 1980s and 1990s, when the field’s canonical examples were meteorological forecasting and direct numerical simulation of turbulence. *Computational Études* brings the subject into the present: it cites work from 2024–2026 throughout, includes applications to quasinormal modes of black holes and wormholes, discusses the interface with Physics-Informed Neural Networks and Fourier Neural Operators, and addresses modern algorithmic developments such as ultraspherical sparse operators, exponential time differencing, and parallel-in-time integration. At the same time, it never loses sight of the classical foundations, and a student who has mastered this book will be as comfortable reading Gottlieb and Orszag’s 1977 monograph as they are reading a 2025 paper on gradient-flow structure-preserving schemes.

The state of the literature on spectral methods is, frankly, uneven. The canonical references are either highly advanced (Canuto et al.’s four-volume series, Shen–Tang–Wang) or narrowly focused (Trefethen’s *Spectral Methods in MATLAB*, indispensable but brief). There has been a genuine gap for a self-contained graduate text that combines theoretical rigour, computational completeness, and modern applications in a single volume. *Computational Études* fills that gap. It will be of immediate value to graduate students in applied mathematics, computational physics, and engineering who are encountering spectral methods for the first time, and it offers enough depth and breadth — particularly in its literature surveys and the starred advanced sections — to be a useful reference for experienced practitioners as well. It is, in short, an invaluable contribution to both science education and the broader research community.

A Possible Mathematical Error

§7.5.2 — Bernstein ellipse parameter for the Witch of Agnesi

Location: Section 7.5.2, Discussion paragraph, first sentence.

Text as printed:

“The Witch of Agnesi $u(x) = 1/(1 + 4x^2)$ has poles at $x = \pm i/2$, giving a Bernstein ellipse parameter $\rho \approx 1 + 1/2 = 1.5$. Accordingly, the error in Table 7 decreases by roughly a factor of 1.5 with each unit increase in N .”

The possible error. The Bernstein ellipse parameter ρ for a singularity at $z_0 = iy$ is given by the standard formula (which the book itself derives correctly in §4.7.2 for the Runge function):

$$\rho = y + \sqrt{y^2 + 1}.$$

For $y = 1/2$:

$$\rho = \frac{1}{2} + \sqrt{\frac{1}{4} + 1} = \frac{1}{2} + \frac{\sqrt{5}}{2} = \frac{1 + \sqrt{5}}{2} \approx 1.618.$$

This is the **golden ratio** φ , not 1.5. The approximation $\rho \approx 1 + y$ is only accurate for singularities very close to the real axis; at $y = 1/2$ it underestimates ρ by approximately 7%.

The subsequent claim that “the error decreases by roughly a factor of 1.5 with each unit increase in N ” is consequently also incorrect.

Numerical check. Table 7 shows errors at $N = 40$ and $N = 50$ of 2.5×10^{-7} and 5.8×10^{-9} respectively, a ratio of ≈ 43 over ten steps, consistent with $1.618^{10} \approx 122$ at leading order (additional factors from the slowly-varying prefactor moderate the raw ratio). The four-step ratios in the denser block ($N = 20 \rightarrow 24 \rightarrow 28 \rightarrow \dots$) are near 5–6, closer to $1.618^4 \approx 6.85$ than to $1.5^4 = 5.06$.

This is also a pleasant didactic observation: the golden ratio appears naturally from the Bernstein ellipse formula applied to the Witch of Agnesi.

Significant Issues — Internal Inconsistencies

§1.6 — Julia omitted from the “Modern Workflow” description

Location: Section 1.6, “A Modern Workflow,” paragraphs 2–3.

Text as printed:

“We utilize a **dual-language approach**:

- Python: For accessibility and integration with the vast open-source scientific ecosystem.
- Matlab: For its historical significance in this field ...”

The problem. The Preface (p. 3) explicitly states:

“We provide implementations in three languages — Python, MATLAB, and Julia — allowing readers to choose their preferred environment.”

The Summary (§1.8, item 5) likewise reads: “... in Python, MATLAB, and Julia, embodying the principle that code and data are integral to the scientific record.” Throughout all 17 chapters, Julia is a full and equal partner with complete implementations in every code block. Describing the approach as “dual-language” in §1.6 while omitting Julia is a substantial inconsistency that will confuse attentive readers.

Suggested fix: Update §1.6 to “triple-language approach” and add Julia as a third bullet, consistent with the Preface and §1.8.

Section-heading capitalisation — inconsistent across chapters

Location: Every chapter’s “Literature Overview” section heading.

Chapters 1–10 use sentence case: “**A non-exhaustive literature overview.**”

Chapters 11–16 switch to title case: “**A Non-Exhaustive Literature Overview.**”

Chapter 17 reverts to sentence case.

The inconsistency is visible both in the Table of Contents and in the body headings. A single uniform style should be applied throughout. Given the book’s overall typographic elegance, sentence case is slightly more consistent with the remaining heading hierarchy, but either convention is acceptable provided it is uniform.

§4.11 vs. §8.8.1 — conflicting attribution for Padua points

Location: Section 4.11 (literature overview) vs. Section 8.8.1 (Padua points remark).

- §4.11 credits “Bos, Caliari, De Marchi, Vianello, and Xu [36, 37]” with *discovering* Padua points.
- §8.8.1 correctly credits “De Marchi, Caliari, and Vianello [176]” as the discoverers.

References [36, 37] are the follow-up papers proving the $\mathcal{O}(\log^2 n)$ Lebesgue constant bound and the Lissajous-curve construction, not the original discovery. The §4.11 phrasing implies the five-author group discovered the points; §8.8.1 is correct.

Suggested fix in §4.11: Attribute the discovery to De Marchi, Caliari, and Vianello [176], then credit Bos et al. [36, 37] for the subsequent Lebesgue constant analysis.

Structural Issues — Table of Contents Gaps

Computational Étude 5.2 absent from the TOC

Location: TOC for Chapter 5; body §5.6.2.

The TOC lists:

- §5.5 — Computational Étude 5.1: The Rational Trigonometric Test p. 104
- §5.7 — Computational Étude 5.3: The Quantum Harmonic Oscillator p. 110

Étude 5.2 (“Higher-Order Derivatives”) exists in the body as §5.6.2 but does not appear in the TOC, making the jump from Étude 5.1 to Étude 5.3 look like a numbering error.

Suggested fix: Promote §5.6.2 to a standalone numbered section, or add a TOC entry for it as a subsection of §5.6.

Computational Étude 10.2 absent from the TOC

Location: TOC for Chapter 10; body §10.2.3.

The TOC shows:

- §10.1 — Computational Étude 10.1: Chebyshev–Fourier Recap p. 217
- §10.4 — Computational Étude 10.3: One-Dimensional Wave Equation p. 223

Étude 10.2 (“Chebyshev FFT Accuracy,” body §10.2.3) is again invisible in the TOC. The same structural pattern applies: études embedded as sub-subsections are not surfaced in the TOC while standalone section-level études are. A brief note in the Preface explaining this convention would set reader expectations; alternatively, all études should receive consistent TOC visibility.

Typos and Minor Issues

Tow instances of “etude” I found without the accent

- §13.5.4, Discussion: “This **etude** also demonstrates a key advantage of Method II. . .”
- One further instance confirmed in the body of Chapter 14.

Every other use of the word in all 17 chapters seems that correctly carries the accent (*étude*), however I would recommend checking again.

Exercise 16.4 — editorial self-correction left in the typeset text

Location: §16.22, Exercise 16.4, part (a).

Text as printed:

“Compute the Fourier interpolant f_N at $N = 16$ Chebyshev-extreme points
(no, equispaced points: this is a periodic function, use equispaced)
and measure...”

The parenthetical is an unremoved editing note. The final text should simply read “at $N = 16$ equispaced points.”

Table 2 — minor numerical discrepancy in §3.3.2

The body text states $u_{\text{exact}}(0) \approx 0.1834$, while Table 2 lists the exact value as 0.1835. The correct value to four decimal places is

$$\frac{1}{4}(1 - \operatorname{sech}(2)) = \frac{1}{4}(1 - 0.26580) \approx 0.18355,$$

which rounds to 0.1836 (or 0.1835 with floor rounding). The text value 0.1834 is off by one unit in the last digit; Table 2’s value 0.1835 is closer. The two should be made consistent.

Figure 3 caption — period formula ambiguity

The caption states “The string oscillates with period $T = 2\pi$.” This is correct for the demo parameters ($L = \pi$, $c = 1$) but the general formula derived in §2.3 is $T = 2L/c$. A parenthetical “(with $L = \pi$, $c = 1$ in this demonstration)” would prevent confusion when readers apply the formula to different parameter choices.

Observations and Suggestions (Non-Errors)

1. **“We” vs. single author.** The Preface uses the academic “we” throughout (“We have written primarily for graduate students...”) but the book has a single author and the Acknowledgements section uses first-person singular. Academic “we” is standard and acceptable; a brief note in the Preface explaining the convention, or a consistent shift to first-person singular, would avoid potential confusion.
2. **Bernstein ellipse formula cross-reference.** The formula $\rho = |z_0 + \sqrt{z_0^2 - 1}|$ is applied in §4.7.2 but is not given a numbered equation or a forward cross-reference. By §7.5.2, when it is needed again, neither the formula nor the §4.7.2 derivation is cited. Adding a numbered equation in §4.7.2 and a “see Eq. (...)” pointer in §7.5.2 would aid readers (and would have prevented the numerical error described above).
3. **Chapter 3 title “Mise en Bouche.”** The French culinary metaphor is elegant and fits the book’s aesthetic. A parenthetical English gloss — “(Appetizer)” or “(A First Taste)” — in the TOC would help non-Francophone readers understand the chapter’s role before opening it.
4. **Missing TOC entry for §2.9, “Exercises.”** The Chapter 2 Exercises section does not appear in the TOC, while every other chapter’s Exercises section is listed. This appears to be a Typst compilation glitch rather than an intentional omission.

Conclusion

The book is mathematically sound throughout the vast majority of its 535 pages. All issues I managed to detect are either structural (TOC gaps, heading capitalisation inconsistencies) or editorial (missed accent on *étude*, one uncleaned in-line self-correction, a small rounding discrepancy in one table).

I believe one last proof reading by the author would bring the manuscript to publication-ready standard. Overall, *Computational Études* is an impressive and pedagogically excellent contribution to the spectral-methods literature and is warmly recommended.
